KAVACH

कवच — the digital armour for zero-harm industrial operations

AI-Powered Industrial Safety Intelligence for Zero-Harm Operations · Problem Statement 1

It sees the compound risk no single alarm can.

Team SlideHustlers · Satyam Kumar · ET AI Hackathon 2.0 (Economic Times × Unstop)

THE PROBLEM

Every safety system worked. Eight workers died.

January 2025, Visakhapatnam Steel Plant. Entrapped gases exploded in a coke oven battery. The facility had functioning gas detectors, digital permit-to-work, SCADA.

The investigation that followed found warning signals in the gas-pressure data — but *“no intelligence layer connected those readings to operational decisions in time.”*

This is the recurring pattern across Indian heavy industry: the data exists, spread across five systems that do not talk. The judgment connecting them lives in one overloaded human head.

6,500+

fatal workplace accidents in India, FY2023 (DGFASLI) — excluding most mining & construction

60%+

of large facilities coordinate their own digital safety tools with manual handoffs (FICCI 2024)

8 lives

lost in a plant where every individual safety system was working as designed

Data present. Unacted upon. That is not a hardware gap — it is a software problem.

WHY IT KEEPS HAPPENING

Each system watches one signal. The combination kills.

DCS / SCADA

alarms when ONE tag crosses ONE limit

Gas detectors

each knows its own ppm, nothing else

e-Permit to work

forms and signatures; blind to live process

CMMS / maintenance

work orders; blind to who is where

Shift handover

one paragraph, written by a tired human

06:30 — the combination no system could see: gas-main pressure drifting upward for 4 hours (below every alarm limit) + crew of five entering the connected basement + gas-main isolation omitted + a single-point gas test + a handover note that missed the drift.

Why can't the plant just wire them together? Trip systems are deliberately simple — certification demands it; nobody certifies judgment. Different vendors, different decades, different departments. The only combination-watcher today is a supervisor at 6 AM, after a week of 200 acknowledged alarms.

THE SOLUTION

KAVACH — the intelligence layer plants are missing

1 FUSE

One live picture from systems that never talked: sensor trends, active permits & isolations, gas tests, work orders, shift changeovers, plant connectivity.

→

2 REASON

Eight compound-risk rules watch combinations, not thresholds — sub-alarm drift + occupied confined space + omitted isolation = CRITICAL, hours before any single alarm.

→

3 ACT

Alert with full evidence and the governing regulation → orchestrated response (suspend permits, isolate, notify, evacuate) → auditable prevention report.

Reads from the systems a plant already owns. Replaces none of them. Safety-instrumented trips stay untouched underneath — KAVACH is the judgment layer above, in software, retrofittable to any plant that has data.

THE DEMO SCENARIO

One morning, replayed minute by minute

- **02:30** — Gas-main pressure starts a slow sub-alarm drift
- **06:00** — Shift handover — the drift is not mentioned
- **06:15** — Gas test: PASS — but single-point only
- **06:30** — Crew of 5 enters basement — KAVACH: CRITICAL
- **08:30** — Hot-work permit 20 m from basement vent
- **09:30** — Riser valve throttled — back-pressure builds
- **09:39** — First uncleared conventional alarm (CO 52 ppm)
- **10:30** — The explosion — in the recorded event

Six individually-explainable signals, one lethal combination. A fictional composite built from publicly reported investigations — every beat is a documented failure mode. All data is synthetic, generated by our deterministic digital twin (34 tagged instruments, 600 scripted minutes, labelled ground truth).

THE PRODUCT

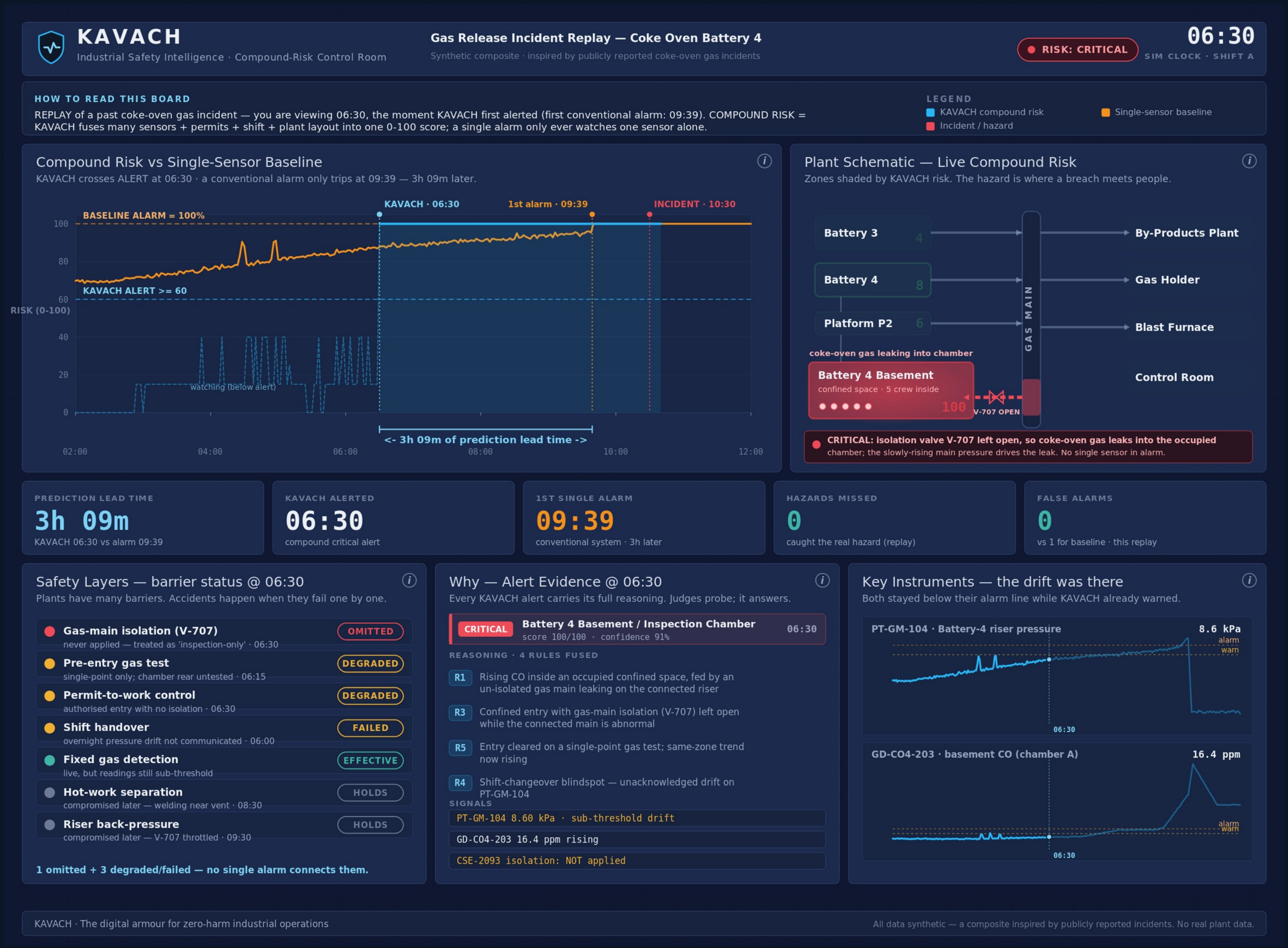
One toggle shows what the plant could not see

BASELINE view

the plant's own single-sensor alarms, replayed faithfully. The board stays green until 09:39 — 50 minutes before the explosion.

KAVACH view

same morning, same data: CRITICAL at 06:30 with four named rules, the driving sensors ringed on the schematic, gas-flow into the occupied basement animated, and every piece of evidence one click away.



MEASURED, NOT CLAIMED

The judges' own evaluation focus — answered with numbers

+3 h 09 m

prediction lead time: compound alert 06:30 vs first uncleared single-sensor alarm 09:39 (189 min)

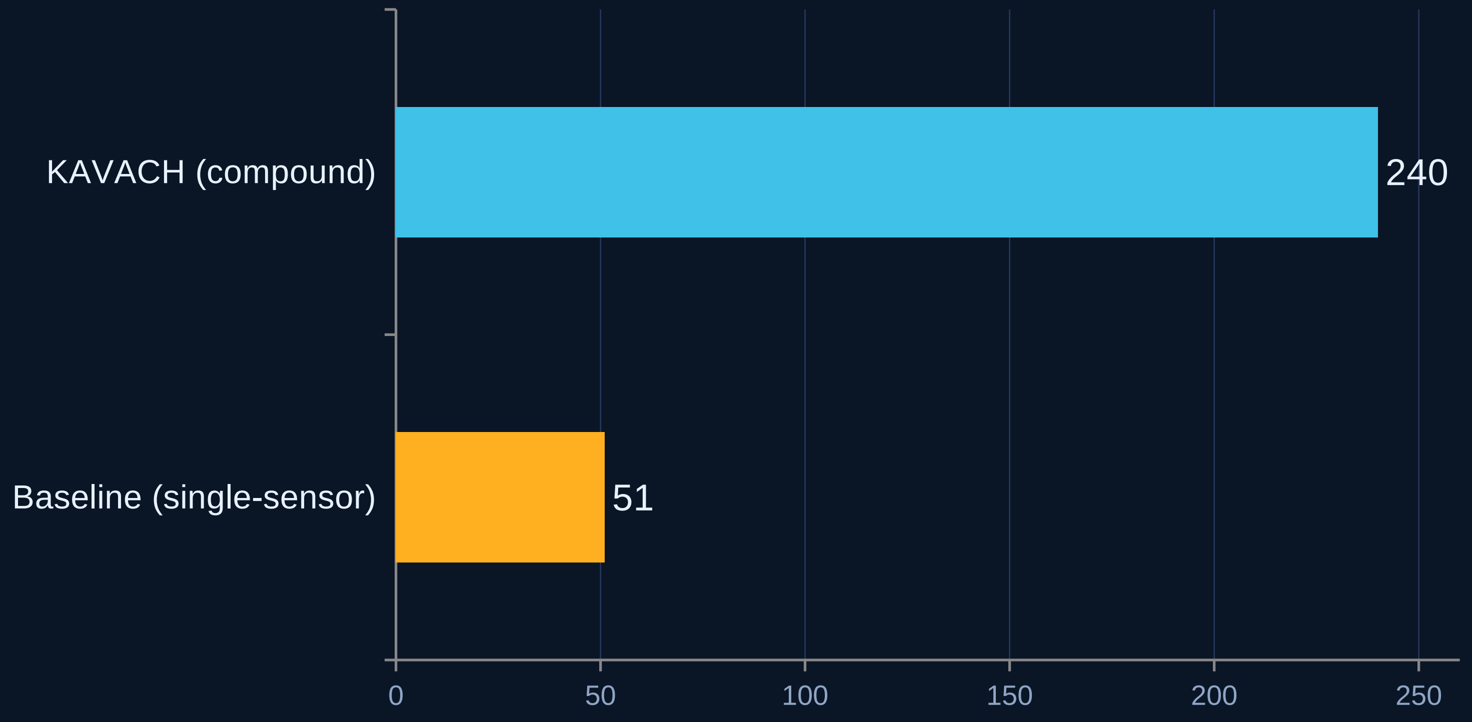
78.8% → 0%

false-negative rate: share of the 240 min of worker exposure left unwarned — baseline vs KAVACH

0 vs 1 · 9 supp.

false alerts on a fully benign day: KAVACH 0, baseline 1 (calibration spike) — 9 context suppressions

Minutes of warning before the incident threshold



Every number is computed by the built-in metrics lab against scenario ground truth, exposed live at /api/metrics, and asserted by the automated verification suite on every run.

BUT WE WROTE THAT SCENARIO

So we tested it on 120 it had never seen

The fair objection

We authored the plant, the scenario, the ground truth and the baseline. A demo you wrote yourself is not evidence.

What we did

A generator produces scenarios the rules have never seen — randomised hazard zone, which barriers fail, drift rate, timings, noise, and a third of them fully benign. The unmodified engine is scored across all of them.

ACROSS 120 UNSEEN SCENARIOS

	KAVACH	baseline
Hazards detected	**100%**	100%
Median lead time	**+85 min**	—
Median FN rate	**0.0%**	39.4%
False alerts, benign days	**0.0%**	0.0%
Warned earlier	**67 of 68**	—

It failed the first time — and that is the point. The held-out run exposed two real defects: 4 missed hazards in a zone the gas rule had overfitted away from, and a 13.5% false-alert rate on benign days (it alerted on CO at 13 ppm against a limit of 30, with the crew fully protected). Both were fixed at the root — multi-hop gas reachability, and crediting intact barriers.

After the fixes: detection 94.1% → 100%, false alerts 13.5% → 0%, median lead 78 → 85 min. The one scenario where KAVACH still loses to the baseline is printed by the tool, not hidden.

EXPLAINABLE BY DESIGN

Every alert arrives with its proof

CRITICAL 100/100 — Battery 4 Basement · 06:30

Rules fired

R1 Rising gas-main trend feeding an occupied confined space
R3 Gas-main isolation OMITTED on the entry permit
R5 Gas test was single-point — chamber rear untested
R4 Shift handover blind spot — drift unacknowledged

Signals, with values

PT-GM-104 8.6 kPa, rising ~4 h (still below the 9.0 warn limit)
Drift detector: sustained +slope while sub-threshold
Connectivity: gm_corridor → cob4_basement via open V-707

Operational context

CSE-2093 — crew of 5 inside · WO-8812 in progress · confidence 0.9

Regulatory basis — attached to the alert

Factories Act 1948, S.36(2)
entry into confined space with dangerous fumes — conditions & certification

OISD-STD-105 — Work Permit System
confined-space / vessel-entry permit: isolation & representative gas testing

DGMS guidance
gas testing by competent persons; continuous monitoring

Not a black box — a witness. Every alert and every orchestrated action carries rules, live values, permits and citations. Audit-ready in a statutory environment.

BEYOND DETECTION

From alert to orchestrated response, with citations

1	2	3	4	5	6	7
Raise	**Suspend**	**Isolate**	**Notify**	**Evacuate**	**Re-test**	**Monitor**
CRITICAL alert with evidence bundle	CSE-2093 confined-space entry halted	close gas-main leg feeding the chamber	shift supervisor · safety officer · control room	zones from the plant connectivity graph	representative multi-point gas test	hold until trend clears and permit re-issued

The counterfactual, computed

Intervention at 06:30 is **240 minutes before the incident** — the crew leaves the chamber 3 hours before conditions turn lethal. The 08:30 hot-work permit and the 09:30 valve throttle **simply never happen.**

Auto-generated Incident-Prevention Report

One click → a regulator-ready document: full timeline, evidence, rules, statutory citations, and the with/without-KAVACH comparison. Markdown + branded PDF, generated by the system itself.

PS asks to compress “the critical first 10 minutes from chaos to coordinated response.” This is that, pre-armed 4 hours early.

STRESS-TEST IT LIVE

Judges don't have to trust us — they can try to break it

“What if the isolation had been applied?”

R3 clears instantly. Zone risk falls 100 → 70, CRITICAL → ALERT. The system rewards the correct barrier.

“What if the gas test had been done properly?”

Representative multi-point test → R5 clears. Score drops further — two barriers restored, risk de-escalates honestly.

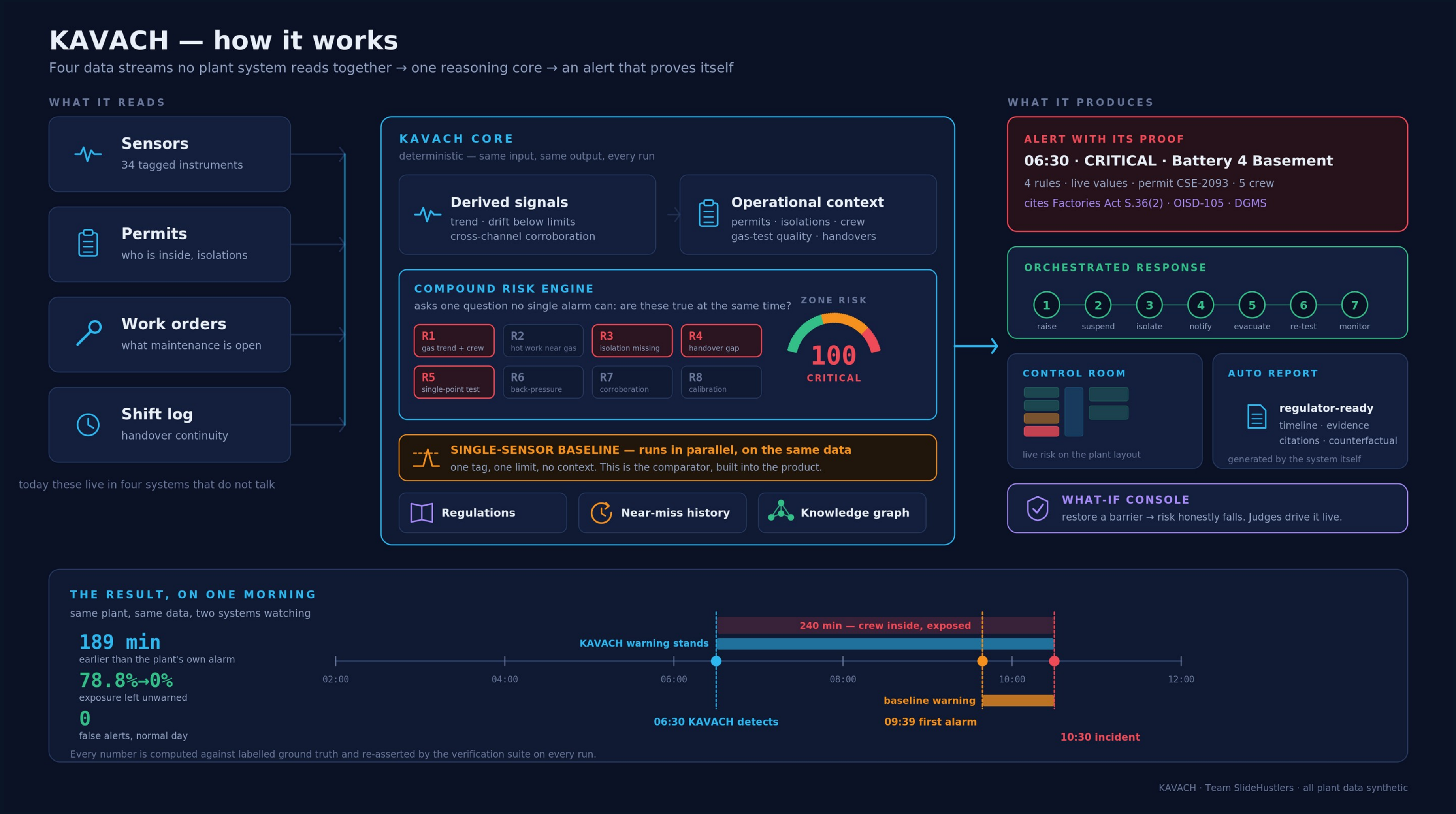
“What if a second CO channel starts rising?”

Corroboration rule R7 joins, confidence rises. More evidence = stronger alert, not alarm spam.

A live sandbox console (/whatif) injects conditions and re-runs the full engine deterministically — same rules, same code path as the replay. **In finale Q&A, the jury drives. Verified: what-if results are bit-identical on recompute.**

UNDER THE HOOD

Deterministic core. Agentic edges.



Stack

FastAPI · pure-Python engines
Next.js 14 · inline-SVG UI (~91 kB)
WebSocket live streaming (4 Hz)

Deterministic

Seeded twin · cached horizon · zero external dependencies — the demo cannot fail

LLM: optional

OpenAI narration of alerts with a hard deterministic fallback

Verified

Automated suite asserts every claim: lead time, FN/FP, rule timings, citations, what-if — ALL PASS

BUSINESS IMPACT

Retrofit, not replace — that is the entire go-to-market

Phase 1 — Observe

Read-only connectors: OPC-UA / historian, e-PTW & CMMS exports. No process writes, no downtime, no re-certification. Weeks, not years.

Phase 2 — Advise

KAVACH alerts with evidence in the control room. Trust is earned on the plant's own near-misses — and measured (FN/FP dashboards).

Phase 3 — Orchestrate

Pre-approved response playbooks: permit suspension, notifications, evacuation sequencing. Human gates above defined thresholds.

Steel · coke · refining · chemicals · mining

asset-heavy sectors where the Vizag pattern repeats — and where Factories Act / OISD / DGMS audits already create compliance pull

New plant = a config, not a project

layout JSON + thresholds + connector mapping; the rules and engines are plant-agnostic — that is the scalability story

The twin ships with the product

operator training, HAZOP-style what-if drills, and regression-testing of rule changes — revenue before the first connector

The ask: one coke-oven battery, 90 days, advisory mode. KAVACH proves itself on the plant's own data — every near-miss it catches is evidence no slide can fake.

Synthetic by necessity — deterministic by design

Our data honesty policy

Real SCADA + permit data is locked behind NDAs — for every team.
We built a high-fidelity digital twin instead:

34 realistically-tagged instruments with physics-consistent behaviour
A 600-minute composite scenario of documented failure modes
A benign day that tests restraint, not just detection
Labelled ground truth, so every metric is honest

Every surface — UI, README, this deck — says the data is synthetic.
Judges should trust what a team says about its own data.

CCTV / PPE vision module

the PS's computer-vision lane — pre-trained detection feeding the same evidence model

Worker positioning (UWB/BLE)

live crew locations on the heatmap; evacuation routing per person

Pattern intelligence

rules learned from near-miss corpora — the PS's incident-pattern lane, seeded by our RAG layer

Fleet view

multi-plant roll-up for corporate EHS — same engines, one pane

Roadmap — each item extends the working evidence model, none re-architects it.

The data already exists in every plant.

KAVACH makes it act in time.

KAVACH · कवच

“Railways built KAVACH for trains. We built it for the workers inside the plants.”